Complex Function Theory 2 - Final Project
Riemann's famous paper on the Zeta function - "Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse", or "On the Number of Primes Less Than a Given Magnitude"

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Spring 2025

Contents

Figure 1: Georg Friedrich Bernhard Riemann (17 September 1826 – 20 July 1866)

1 Background - Riemann

Georg Friedrich Bernhard Riemann was born on 17 September 1826 in Breselenz, a village near Dannenberg in the Kingdom of Hanover (German Confederation). His father, Friedrich Bernhard Riemann, was a poor Lutheran pastor in Breselenz who fought in the Napoleonic Wars (died in 1855). His mother, Charlotte Ebell, died in 1846. Riemann was the second of six children. Riemann exhibited exceptional mathematical talent, such as calculation abilities, from an early age but suffered from timidity and a fear of speaking in public, and had frail health.

Riemann studied the Bible intensively, but he was often distracted by mathematics. His teachers were amazed by his ability to perform complicated mathematical operations, in which he often outstripped his instructor's knowledge. In 1846, at the age of 19, he started studying philology and Christian theology in order to become a pastor and help with his family's finances.

During the spring of 1846, his father, after gathering enough money, sent Riemann to the University of Göttingen, where he planned to study towards a degree in theology. However, once there, he began studying mathematics under Carl Friedrich Gauss. Gauss recommended that Riemann give up his theological work and enter the mathematical field; after getting his father's approval, Riemann transferred to the University of Berlin in 1847.

Riemann was a dedicated Christian, the son of a Protestant minister, and saw his life as a mathematician as another way to serve God. He died of tuberculosis on 20 July 1866. At the time of his death, he was reciting the "Lord's Prayer" with his wife and died before they finished saying the prayer. Four of Bernhard's five siblings died unexpectedly one after another 10 years prior to his death.

2 Background - The Paper

The theory of which Riemann's paper is a part had its beginnings in Euler's theorem, proved in 1737, that the sum of the reciprocals of the prime numbers diverges:

$$\sum_{p \text{ prime}} \frac{1}{p} = \frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{11} + \frac{1}{13} + ... = \infty$$

Euler in fact says that since $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + ...$ diverges like the logarithm and since $\sum_p \frac{1}{p}$ diverges like the logarithm of $\sum_n \frac{1}{n}$, the series $\sum_p \frac{1}{p}$ must diverge as the log of the log (which he writes as $\log(\log \infty)$). Thus if μ is the point measure that assigns the value 1 to primes and 0 to all other points:

$$\int_2^x \frac{d\mu}{v} = \sum_{p<x} \frac{1}{p} \sim \log(\log(x)) = \int_1^{\log x} \frac{du}{u} = \int_e^x \frac{dv}{v \log v}$$

So we get heuristically that the density of primes is roughly $\frac{1}{\log x}$, which is the content of the Prime Number Theorem. In his paper, Riemann laid the foundations of the methods and ideas that ultimately led to its proof about 40 years later.

"On the Number of Primes Less Than a Given Magnitude" is a seminal 9-page paper by Riemann published in the November 1859 edition of the Monatsberichte der Königlich Preußischen Akademie der Wissenschaften zu Berlin.

This paper studies the prime-counting function using analytic methods. Although it is the only paper Riemann ever published on number theory, it contains ideas which influenced thousands of researchers during the late 19th century and up to the present day. The paper consists primarily of definitions, heuristic arguments, sketches of proofs, and the application of powerful analytic methods; all of these have become essential concepts and tools of modern analytic number theory.

Among the new definitions, ideas, and notation introduced:

- The use of the Greek letter zeta (ζ) for a function previously mentioned by Euler
- The analytic continuation of this zeta function $\zeta(s)$ to all complex $s \neq 1$

- The entire function $\xi(s)$, related to the zeta function through the gamma function Γ
- The discrete function $J(x)$ defined for $x \geq 0$, which is defined by $J(0) = 0$ and $J(x)$ jumps by $\frac{1}{n}$ at each prime power p^n

Among the proofs and sketches of proofs:

- Two proofs of the functional equation of $\zeta(s)$
- Proof sketch of the product representation of $\xi(s)$
- Proof sketch of the approximation of the number of roots of $\xi(s)$ whose imaginary parts lie between 0 and T
- The main result of the paper: a formula for $J(x)$, by comparing with $\ln(\zeta(s))$. Riemann then found a formula for the prime-counting function $\pi(x)$
- Formulation of his famous "Riemann Hypothesis"

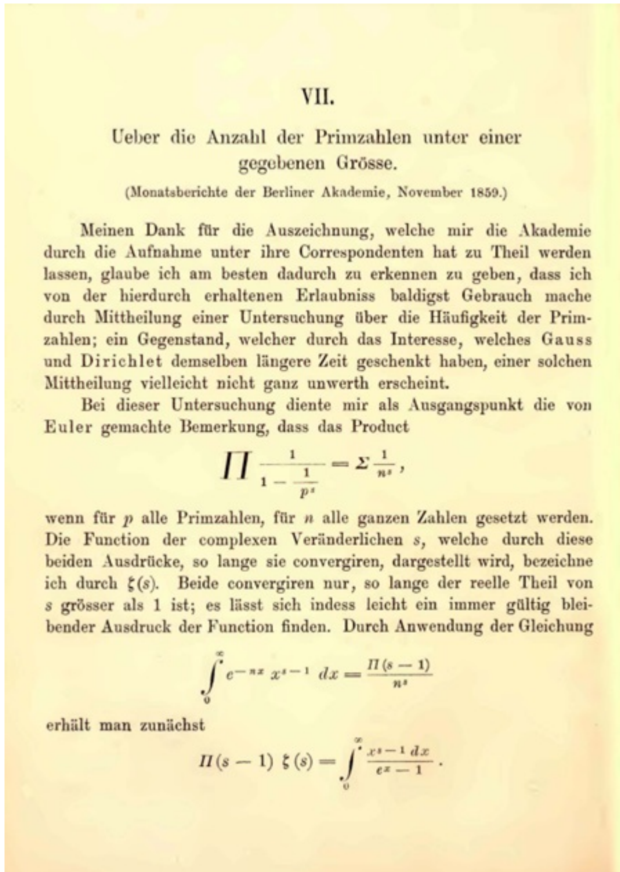
The paper contains some peculiarities for modern readers, such as the use of $\Pi(s-1)$ instead of $\Gamma(s)$, writing tt instead of t^2, and using the bounds of ∞ to ∞ to denote a contour integral.

A brief outline of Riemann's paper:

1. He begins with Euler's product representation for $\zeta(s)$, but notes that "They converge only when the real part of s is greater than 1"
2. Then he sketches a theory of the zeta function in which he:
   (a) Extends the domain of ζ to include all complex numbers s except $s = 1$
   (b) Prove a functional equation and symmetric formula for the Zeta function (for the later he gives to proofs)
   (c) Proves, states and conjectures certain results about ξ's zeros (including the Riemann Hypothesis)
   (d) Obtains a product representation of $\xi(s)$ in terms of its zeros
3. He obtains $\log(\zeta(s))$ as an integral involving $J(x)$
4. Uses Fourier inversion to express $J(x)$ as an integral involving $\ln(\zeta(s))$
5. Evaluates the integral to obtain a formula for $J(x)$
6. Uses Möbius inversion to obtain a formula for $\pi(x)$ in terms of $J(x)$

7. Proposes an improved approximation to $\pi(x)$ in terms of $\mathrm{Li}(x)$ prime powers

Throughout this paper, we draw on the works of Edwards (2001), Riemann (1876), and Wilkins (1998) translation of Riemann.



VII.

Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse.

(Monatsberichte der Berliner Akademie, November 1859.)

Meinen Dank für die Auszeichnung, welche mir die Akademie durch die Aufnahme unter ihre Correspondenten hat zu Theil werden lassen, glaube ich am besten dadurch zu erkennen zu geben, dass ich von der hierdurch erhaltenen Erlaubniss baldigst Gebrauch mache durch Mittheilung einer Untersuchung über die Häufigkeit der Primzahlen; ein Gegenstand, welcher durch das Interesse, welches Gauss und Dirichlet demselben längere Zeit geschenkt haben, einer solchen Mittheilung vielleicht nicht ganz unwerth erscheint.

Bei dieser Untersuchung diente mir als Ausgangspunkt die von Euler gemachte Bemerkung, dass das Product

$$\Pi \frac{1}{1-\frac{1}{p^s}} = \Sigma \frac{1}{n^s},$$

wenn für p alle Primzahlen, für n alle ganzen Zahlen gesetzt werden. Die Function der complexen Veränderlichen s, welche durch diese beiden Ausdrücke, so lange sie convergiren, dargestellt wird, bezeichne ich durch $\zeta(s)$. Beide convergiren nur, so lange der reelle Theil von s grösser als 1 ist; es lässt sich indess leicht ein immer gültig bleibender Ausdruck der Function finden. Durch Anwendung der Gleichung

$$\int_0^\infty e^{-nx} x^{s-1}\, dx = \frac{\Pi(s-1)}{n^s}$$

erhält man zunächst

$$\Pi(s-1)\,\zeta(s) = \int_0^\infty \frac{x^{s-1}\,dx}{e^x-1}.$$

Figure 2: Riemann's original article, page 1

3 Zeta function - Analytic Continuation

Riemann's paper begins with the celebrated Euler product formula

$$\prod_p \left(1 - \frac{1}{p^s}\right)^{-1} = \sum \frac{1}{n^s} = \zeta(s).$$

He then proves the identity

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx,$$

and uses it to extend the ζ function to the entire plane as follows (See exercise 34):

First we compute for $x > 0$ (when $|e^{-x}| < 1$):

$$\frac{x^{s-1}}{e^x - 1} = \frac{x^{s-1}e^{-x}}{1 - e^{-x}} = x^{s-1}e^{-x}\sum_{n=0}^{\infty} e^{-xn} = x^{s-1}\sum_{n=1}^{\infty} e^{-xn} = \sum_{n=1}^{\infty}(nx)^{s-1}e^{-xn}\frac{1}{n^{s-1}}.$$

Now for $\Re(s) > 1$ we have

$$\int_0^{\infty} \frac{x^{s-1}}{e^x - 1}dx = \int_0^{\infty}\sum_{n=1}^{\infty}(nx)^{s-1}e^{-xn}\frac{1}{n^{s-1}}dx = \sum_{n=1}^{\infty}\frac{1}{n^{s-1}}\int_0^{\infty}(nx)^{s-1}e^{-xn}dx$$
$$= \sum_{n=1}^{\infty}\frac{1}{n^{s-1}}\int_0^{\infty} t^{s-1}e^{-t}\frac{dt}{n} = \sum_{n=1}^{\infty}\frac{1}{n^s}\int_0^{\infty} t^{s-1}e^{-t}dt = \zeta(s)\Gamma(s).$$

Where the change of sum and integral is justified by absolute convergence so Fubin's theorem applies. Dividing by $\Gamma(s)$ (which doesn't vanish) we indeed get

$$\zeta(s) = \frac{1}{\Gamma(s)}\int_0^{\infty}\frac{x^{s-1}}{e^x - 1}dx.$$

Now let us use this to extend the zeta function. Fix some $\varepsilon > 0$ small and look at the following paths

$$\gamma_1 = \{re^{i\varepsilon} \ : \ \varepsilon \le r < \infty\},$$
$$\gamma_2 = \{\varepsilon e^{i\theta} \ : \ \varepsilon \le \theta \le 2\pi - \varepsilon\},$$
$$\gamma_3 = \{re^{-i\varepsilon} \ : \ \varepsilon \le r < \infty\},$$
$$\gamma = \gamma_1 \cup \gamma_2 \cup \gamma_3.$$

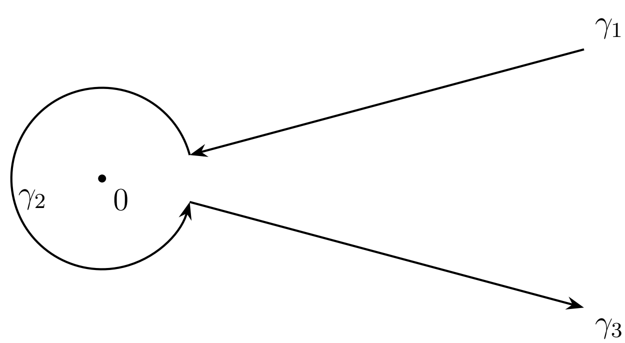


Figure 3: The path γ

Where the path start from infinity, goes to $\varepsilon e^{i\varepsilon}$, then rotates anti clock wise around 0 to

$\varepsilon e^{-i\varepsilon}$, and then back to infinity. We wish to compute the integral:

$$\int_\gamma \frac{(-z)^s}{e^z-1}dz$$

where $(-z)^s = e^{s\log(-z)}$, and we take log to be the principle branch. In particular as z goes to -1 from above the real line $\log z$ goes to $i\pi$ and as it goes to -1 from below the real line $\log z$ goes to $-i\pi$. Hence on γ_1 the integrand is approximately $\frac{e^{-is\pi}r^s}{e^r-1}$ and on γ_3 it is approximately $\frac{e^{is\pi}r^s}{e^r-1}$. Formally we have:

$$\begin{aligned}
&\int_{\gamma_1} \frac{(-z)^s}{e^z-1}dz = \int_\infty^\varepsilon \frac{\exp(s\log(-re^{i\varepsilon}))e^{i\varepsilon}dr}{\exp(re^{i\varepsilon})-1} = -e^{i\varepsilon}\int_\varepsilon^\infty \frac{\exp(s\log(re^{i(\varepsilon-\pi)}))dr}{\exp(re^{i\varepsilon})-1} = \\
&= -\ e^{i(-s\pi+s\varepsilon+\varepsilon)}\int_\varepsilon^\infty \frac{r^s dr}{\exp(re^{i\varepsilon})-1}, \\
&\int_{\gamma_3} \frac{(-z)^s}{e^z-1}dz = \int_\varepsilon^\infty \frac{\exp(s\log(-re^{-i\varepsilon}))e^{-i\varepsilon}dr}{\exp(re^{-i\varepsilon})-1} = e^{-i\varepsilon}\int_\varepsilon^\infty \frac{\exp(s\log(re^{i(\pi-\varepsilon)}))dr}{\exp(re^{-i\varepsilon})-1} = \\
&= e^{i(s\pi-s\varepsilon-\varepsilon)}\int_\varepsilon^\infty \frac{r^s dr}{\exp(re^{i\varepsilon})-1}.
\end{aligned}$$

Now clearly the integral over γ_2 goes to 0 as ε goes to 0 (for $\Re(s) > 2$), so using dominated convergence (again for $\Re(s) > 2$) we get

$$\lim_{\varepsilon\to 0}\int_\gamma \frac{(-z)^s}{e^z-1}dz = \left(e^{i\pi s} - e^{-i\pi s}\right)\int_0^\infty \frac{x^s dx}{e^x-1}.$$

Replacing s with $s-1$ and using $e^{\pm i\pi} = -1$ we get

$$\lim_{\varepsilon\to 0}\int_\gamma \frac{(-z)^{s-1}}{e^z-1}dz = \zeta(s)\Gamma(s)(e^{i\pi(s-1)} - e^{-i\pi(s-1)}) = \zeta(s)\Gamma(s)(e^{-i\pi s} - e^{i\pi s}).$$

Dividing in the above and recalling $\sin z = \frac{e^{iz}-e^{-iz}}{2i}$, we get

$$\zeta(s) = \frac{i}{2\Gamma(s)\sin(\pi s)}\lim_{\varepsilon\to 0}\int_\gamma \frac{(-z)^{s-1}}{e^z-1}dz.$$

Now the above is certainly valid for any $\Re(s) > 1$, but we claim that it is valid for any $s \neq 1$ (where we have a simple pole). Indeed, the integral $\lim_{\varepsilon\to 0}\int_\gamma \frac{(-x)^s}{e^x-1}dx$, always converge since e^x decays faster than x^s as x goes to infinity (and it is holomorphic like we saw in class when talking about poisson kernels using morera's theorem), so we only need to worry about the poles of $\frac{1}{\sin(\pi s)\Gamma(s)}$ (note, this is actually equal $\frac{\Gamma(1-s)}{\pi}$). The poles of this function must all be at the natural numbers; at the nonpositive integers, the zeros of $\sin \pi s$

cancel with the poles of $\Gamma(s)$, and at positive (greater then 1) integers the standard formula $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$, is valid so we can't have a pole (or rather any pole of this function must cancel out with a 0 of the integral) and we don't need this formula for such s anyway. And at $s = 1$ the $\sin(\pi s)$ is zero but Γ isn't so we have a simple pole.

4 Riemann's Functional Equation

4.1 First proof

Riemann gives two proofs for his functional equation formula, first he uses the above formula

$$2\sin(\pi s)\Gamma(s)\zeta(s) = i\int_{\gamma} \frac{(-z)^{s-1}}{e^z - 1} dz$$

Note that as $|z| \to \infty$ the function $\frac{(-z)^{s-1}}{e^z-1}$ goes to 0 exponentially fast provided that $\mathfrak{Re}(s) < 0$, hence if we stop the path at some large value $|z| = r$, and make the path a closed path by going counter clockwise from the 'end' of γ_3 to the 'start' of γ_1 (see figure 4) the value on the outer arc will be negligible, so at the limit $r \to \infty$ the integral on this closed path $\bar{\gamma}$ will be equal to the one on the original path γ.

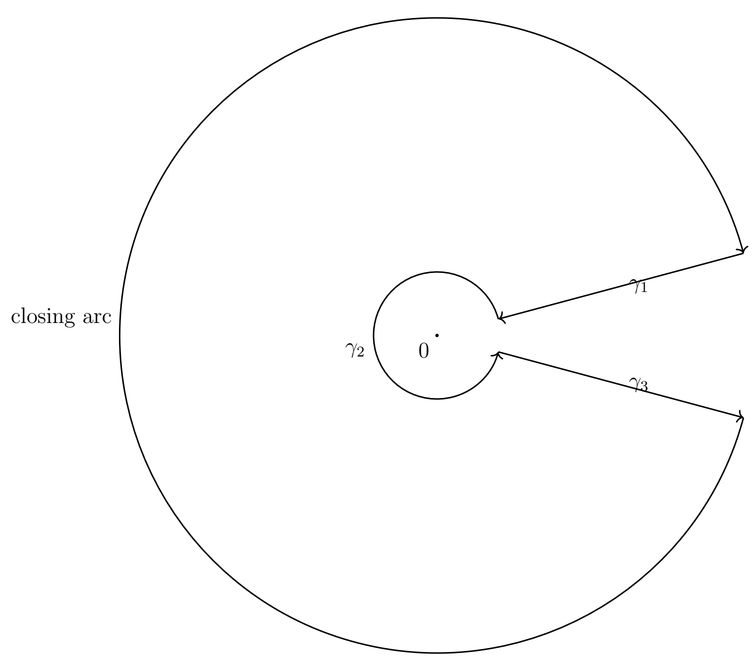


Figure 4: The closed path $\bar{\gamma}$

The advantage of working with a closed loop is that now we can use Cauchy residue theorem to compute the integral. The (simple) poles of $\frac{(-z)^{s-1}}{e^z-1}$ are at the points of the form $z = 2\pi ik$, for k non zero integer (the pole at zero is outside our contour), and the residue there is $(-k2\pi i)^{s-1}$. Hence, by the residue theorem, we get (note the minus sign in the sum of the residues since we took the integral in negative orientation)

$$\begin{aligned}2\sin(\pi s)\Gamma(s)\zeta(s) &= i\int_{\bar{\gamma}} \frac{(-z)^{s-1}}{e^z-1}dz = -i2\pi i\sum_n (-n2\pi i)^{s-1} + (n2\pi i)^{s-1} \\ &= \zeta(1-s)(2\pi)^s[i^{s-1} + (-i)^{s-1}] = \zeta(1-s)(2\pi)^s 2\sin(\frac{\pi s}{2}).\end{aligned}$$

Dividing by $2\sin(\pi s)\Gamma(s)$, and using the identity $\frac{\pi}{\sin(\pi s)\Gamma(s)} = \Gamma(1-s)$, we get for $\mathfrak{Re}(s) < 0$ the famous formula

$$\zeta(s) = 2^s\pi^{s-1}\sin(\frac{\pi s}{2})\Gamma(1-s)\zeta(1-s).$$

So by the identity theorem the equality holds for all s. Riemann then utilizes the following identities of the Gamma function (reflection and duplication formulas)

$$\begin{aligned}\Gamma(1-s)\Gamma(s) &= \frac{\pi}{\sin\pi s}, \\ \Gamma(s)\Gamma(s+\frac{1}{2}) &= 2^{1-2s}\sqrt{\pi}\Gamma(2s).\end{aligned}$$

Those facts together with the functional equation of the Zeta function implies that the function $s \mapsto \Gamma(\frac{s}{2})\pi^{\frac{-s}{2}}\zeta(s)$ is invariant to the transformation $s \mapsto 1-s$.

4.2 Second proof

Riemann gives another proof of his functional equation formula, which goes along the lines of his expansion of the Zeta function. Indeed, starting from the Gamma function and using the change of variable $t = n^2\pi x$, one has for $\mathfrak{Re}(s) > 1$

$$\Gamma(\frac{s}{2}) = \int_0^\infty t^{\frac{s}{2}-1}e^{-t}dt = \int_0^\infty x^{\frac{s}{2}-1}n^{s-2}\pi^{\frac{s}{2}-1}e^{-xn^2\pi}\cdot(n^2\pi)dx \implies \frac{1}{n^s}\pi^{\frac{-s}{2}}\Gamma(\frac{s}{2}) = \int_0^\infty x^{\frac{s}{2}-1}e^{-n^2\pi x}dx.$$

Hence, defining

$$\psi(x) = \sum_n e^{-n^2\pi x},$$

and summing on the above yields

$$\zeta(s)\pi^{\frac{-s}{2}}\Gamma(\frac{s}{2}) = \int_0^\infty x^{\frac{s}{2}-1}\psi(x)dx, \quad \mathfrak{Re}(s) > 1$$

Now Riemann utilizes the following identity of the ψ function, which was proved by Jacobi (using the Poisson summation formula from Fourier theory)

$$2\psi(x) + 1 = \frac{2\psi(\frac{1}{x}) + 1}{\sqrt{x}},$$

from which he deduces

$$\int_0^\infty x^{\frac{s}{2}-1}\psi(x)dx = \int_1^\infty x^{\frac{s}{2}-1}\psi(x)dx - \int_\infty^1 t^{-\frac{s}{2}-1}\psi(\frac{1}{t})dt = \int_1^\infty \frac{\psi(x)}{x}(x^{\frac{1-s}{2}} + x^{\frac{s}{2}})dx + \frac{1}{2}\int_1^\infty \frac{(x^{\frac{s-1}{2}} - x^{\frac{-s}{2}})}{x}dx.$$

The second integral can be evaluated to $\frac{1}{s(s-1)}$ hence we get

$$\zeta(s)\pi^{\frac{-s}{2}}\Gamma(\frac{s}{2}) = \int_1^\infty \frac{\psi(x)}{x}(x^{\frac{1-s}{2}} + x^{\frac{s}{2}})dx - \frac{1}{s(1-s)}$$

This gives another extension of the Zeta function since ψ is rapidly decreasing and the integral converges for all s, thus by the identity theorem the formula holds for all s. Because the right side is obviously invariant to the transformation $s \mapsto 1-s$, this proves the functional equation of the zeta function.

5 ξ function

In view of the above formula, Riemann defines the function

$$\xi(s) = \zeta(s)\pi^{\frac{-s}{2}}\Gamma(\frac{s}{2})\frac{s}{2}(s-1)$$

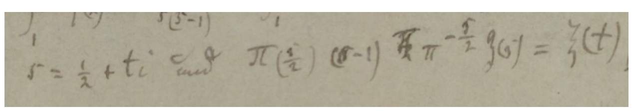


Figure 5: For $s = \frac{1}{2} + ti$, Riemann defines $\xi(t) := \zeta(s)\pi^{\frac{-s}{2}}\Gamma(\frac{s}{2})\frac{s}{2}(s-1)$

Which is invariant under the transformation $s \mapsto 1-s$, then using integration by parts twice he computes:

$$\begin{aligned}
\xi(s) &= \zeta(s)\pi^{\frac{-s}{2}}\Gamma(\frac{s}{2})\frac{s}{2}(s-1) = \frac{1}{2} - \frac{s(1-s)}{2}\int_1^\infty \frac{\psi(x)}{x}(x^{\frac{1-s}{2}} + x^{\frac{s}{2}})dx \\
&= \frac{1}{2} - \frac{s(1-s)}{2}\Big(\psi(x)\big(\frac{2x^{\frac{1-s}{2}}}{1-s} + \frac{2x^{\frac{s}{2}}}{s}\big)\Big)\Big|_1^\infty + \frac{s(1-s)}{2}\int_1^\infty \psi'(x)\big(\frac{2x^{\frac{1-s}{2}}}{1-s} + \frac{2x^{\frac{s}{2}}}{s}\big)dx \\
&= \frac{1}{2} + \psi(1) + \int_1^\infty \psi'(x)(sx^{\frac{1-s}{2}} + (1-s)x^{\frac{s}{2}})dx \\
&= \frac{1}{2} + \psi(1) + \int_1^\infty x^{\frac{3}{2}}\psi'(x)(sx^{\frac{-s}{2}-1} + (1-s)x^{\frac{s-1}{2}-1})dx \\
&= \frac{1}{2} + \psi(1) + \Big(x^{\frac{3}{2}}\psi'(x)\big(-2x^{\frac{-s}{2}} - 2x^{\frac{s-1}{2}}\big)\Big)\Big|_1^\infty + 2\int_1^\infty (x^{\frac{3}{2}}\psi'(x))'[x^{\frac{-s}{2}} + x^{\frac{s-1}{2}}]dx \\
&= \frac{1}{2} + \psi(1) + 4\psi'(1) + 2\int_1^\infty (x^{\frac{3}{2}}\psi'(x))'[x^{\frac{-s}{2}} + x^{\frac{s-1}{2}}]dx.
\end{aligned}$$

Now differentiating the functional equation of $\psi(x)$ will show that $\frac{1}{2} + \psi(1) + 4\psi'(1) = 0$. This with the fact that $x^{\frac{-s}{2}+\frac{1}{4}} + x^{\frac{s-1}{2}+\frac{1}{4}} = e^{\frac{1-2s}{4}\log x} + e^{\frac{2s-1}{4}\log x} = 2\cosh(\frac{2s-1}{4}\log x)$, gives

$$\xi(s) = 4\int_1^\infty (x^{\frac{3}{2}}\psi'(x))'x^{\frac{-1}{4}}\cosh(\frac{1}{2}(s-\frac{1}{2})\log x)dx.$$

Which Riemann writes as[1]

$$\xi(\frac{1}{2}+it) = 4\int_1^\infty (x^{\frac{3}{2}}\psi'(x))'x^{\frac{-1}{4}}\cos(\frac{t}{2}\log x)dx.$$

Riemann remarks that ξ is an analytic function in the entire plane, and that all its zeros lie in the strip of complex numbers with real value between 0 and 1. He also claims that the number of such roots with imaginary part between 0 and iT is approximately

$$\frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi}.$$

He claims that this is a consequence of the fact that the integral of $\frac{\xi'}{\xi}$ on the contour surrounding this domain is approximation for this number (times $2\pi i$). This method of counting zeroes (and poles) is known as the argument principle and is an easy consequence of the residue theorem, for the poles of $\frac{\xi'}{\xi}$ are precisely the roots ξ and the residue in each such pole is 1. While the above estimate is accurate, Riemann gives no insight as to how he evaluated the integral of $\frac{\xi'}{\xi}$, a task which turns out to be non trivial at all. In fact, it

[1]Actually Riemman uses the letter ξ to denote the function which is now customary to denote by Ξ, which is $\Xi(t) := \xi(\frac{1}{2}+it)$ where ξ is defined as above

took until 1905 (over 45 years!) that von Mangoldt succeeded in proving Riemann's claim. Riemann also claims that the number of roots of ξ in this domain with real part $\frac{1}{2}$ is also approximately

$$\frac{T}{2\pi}\log\frac{T}{2\pi}-\frac{T}{2\pi},$$

which led him to conjecture that **all** roots of ξ have real part $\frac{1}{2}$, a conjecture which is now known as the Riemann hypothesis, and is one of the biggest open problems in mathematics. We note that we do not know how Riemann got his claim for the approximation[2] of the number of roots of ξ with real value $\frac{1}{2}$, and even today this assertion remains unproven. Finally, Riemann also claims that the ξ function has an extension as a product

$$\xi(s)=\xi(0)\prod_{\rho}(1-\frac{s}{\rho}),$$

where the product is over all roots of ξ, or taking logarithm

$$\log\xi(s)=\log\xi(0)+\sum_{\rho}\log(1-\frac{s}{\rho}).$$

Riemann justifies this by noting that the sum converge thanks to his earlier estimation of the number of roots up to a given point. We can also relate the above fact with more modern tools such as the Hadmarad factorization theorem.

6 Fourier Inversion

Let $\pi(x)$ be the function equal to the number of prime numbers less than x, then taking log over the Euler product formula one has for $\mathfrak{Re}(s)>1$

$$\log\zeta(s)=\sum_{p}-\log(1-p^{-s})=\sum_{p}\sum_{n}\frac{1}{np^{sn}}=\sum_{p}\sum_{n}\frac{s}{n}\int_{p^n}^{\infty}x^{-s-1}dx=\sum_{n}\frac{s}{n}\int_{1}^{\infty}\pi(x^{\frac{1}{n}})x^{-s-1}dx$$

Hence after denoting

$$J(x)=\sum_{n}\frac{\pi(x^{\frac{1}{n}})}{n},$$

one has

$$\frac{\log\zeta(s)}{s}=\int_{1}^{\infty}J(x)x^{-s-1}dx,\quad \mathfrak{Re}(s)>1$$

[2]Riemann writes "Certainly one would wish for a stricter proof here; I have meanwhile temporarily put aside the search for this after some fleeting futile attempts, as it appears unnecessary for the next objective of my investigation"

From this Riemann uses Fourier inversion to deduce

$$J(x) = \frac{1}{2\pi i}\int_{\sigma-i\infty}^{\sigma+i\infty} \frac{\log\zeta(s)}{s} x^s ds, \quad \sigma > 1$$

His use of inversion techniques was highly innovative for the time, anticipating ideas that Mellin would formalize with his transform nearly four decades later. Riemann writes that if $h(x), g(s)$ are functions such that $h(x)$ is real and

$$g(s) = \int_0^\infty h(x)x^{-s-1}dx,$$

then, for $s = \sigma + it$, putting $\phi(u) = 2\pi h(e^u)e^{-\sigma u}$ and $x = e^u$ gives

$$g(\sigma + it) = \int_0^\infty h(x)x^{-s-1}dx = \int_{-\infty}^\infty h(e^u)e^{-u(\sigma+it)}du = \frac{1}{2\pi}\int_{-\infty}^\infty \phi(u)e^{-iut}du.$$

From which the standard Fourier inversion formula implies

$$2\pi h(e^u)e^{-\sigma u} = \phi(u) = \int_{-\infty}^\infty g(\sigma + it)e^{iut}dt,$$

from which we get

$$h(x) = \frac{1}{2\pi}\int_{-\infty}^\infty g(\sigma + it)x^{\sigma+it}dt = \frac{1}{2\pi i}\int_{\sigma-i\infty}^{\sigma+i\infty} g(s)x^s ds.$$

Applying this to the pair $\frac{\log\zeta(s)}{s}$, $J(x)$ we get that indeed, for any $\sigma > 1$ we have

$$J(x) = \frac{1}{2\pi i}\int_{\sigma-i\infty}^{\sigma+i\infty} \frac{\log\zeta(s)}{s} x^s ds \tag{1}$$

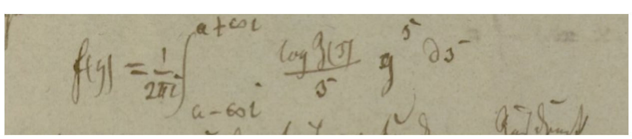


Figure 6: Riemann's formula for $J(x)$ in terms of integral of $\frac{\log\zeta(s)}{s}$

7 Li function

Recall that we have

$$\xi(0)\prod_{\rho}(1-\frac{s}{\rho}) = \xi(s) = \zeta(s)\pi^{\frac{-s}{2}}\Gamma(\frac{s}{2})\frac{s}{2}(s-1) = \zeta(s)\pi^{\frac{-s}{2}}\Gamma(\frac{s}{2}+1)(s-1),$$

hence taking log on both sides we have

$$\log\zeta(s) = \log\xi(0) + \sum_{\rho}\log(1-\frac{s}{\rho}) + \frac{s}{2}\log\pi - \log\Gamma(\frac{s}{2}+1) - \log(s-1)$$

We wish to substitute this into (1), but since the individual terms in the sum will have a non converging integral, first Riemann applies integration by parts[3]:

$$J(x) = \frac{-1}{2\pi i\log x}\int_{\sigma-i\infty}^{\sigma+i\infty}(\frac{\log\zeta(s)}{s})'x^s ds.$$

Let us carefully evaluate the terms. For the $\log\xi(0)$ we may undo the integration by parts and get (from the residue theorem for example)

$$\frac{-1}{2\pi i\log x}\int_{\sigma-i\infty}^{\sigma+i\infty}(\frac{\log\xi(0)}{s})'x^s ds = \frac{1}{2\pi i}\int_{\sigma-i\infty}^{\sigma+i\infty}\frac{\log\xi(0)}{s}x^s ds = \log\xi(0).$$

The third term $\frac{s}{2}\log\pi$ vanishes after dividing by s and differentiating. Now expending $\Gamma(\frac{s}{2}+1)$ into a product and taking log

$$\log\Gamma(\frac{s}{2}+1) = \sum_{n=1}^{\infty}\left(-\log(1+\frac{s}{2n}) + \frac{s}{2}\log(1+\frac{1}{n})\right)$$

we get that the remaining terms are all of the from $\log(1-\frac{s}{\beta})$[4] for various values of β (the right term in the above formula vanishes like the $\frac{s}{2}\log\pi$ term). Riemann approach is as follows: define $F(\beta,x)$ by

$$F(\beta,x) := \frac{1}{2\pi i\log x}\int_{\sigma-i\infty}^{\sigma+i\infty}(\frac{d}{ds}\cdot\frac{\log(1-\frac{s}{\beta})}{s})x^s ds$$

[3]The non-integral term vanishes because $\lim_{t\to\infty}\frac{\zeta(\sigma\pm it)}{\sigma\pm it}x^{\sigma\pm it} = 0$, because $|\log\zeta(a\pm it)| = |\sum_{n,p}(\frac{1}{n}p^{-n(\sigma\pm it)})| \le |\sum_{n,p}(\frac{1}{n}p^{-n\sigma})| = \log\zeta(\sigma) = const$, and the denominator goes to infinity

[4]$\log(s-1)$ is slightly different but the same approach works for it

differentiating under the integral sign, swapping the order of differentiation and integrating by parts gives (provided $\sigma > \mathfrak{Re}(\beta)$)

$$\frac{d}{d\beta}F(\beta,x) = \frac{1}{2\pi i \log x}\int_{\sigma-i\infty}^{\sigma+i\infty}\left(\frac{d}{ds}\cdot\frac{1}{\beta(\beta-s)}\right)x^s ds = \frac{-1}{2\pi i}\int_{\sigma-i\infty}^{\sigma+i\infty}\frac{1}{\beta(\beta-s)}x^s ds = \frac{x^\beta}{\beta},$$

where the last equality follows by applying Fourier inversion or using the residue theorem. Thus we get

$$\frac{d}{d\beta}F(x,\beta) = \frac{x^\beta}{\beta} = \int_A^x y^{\beta-1}dy,$$

where $A = 0$ if $\mathfrak{Re}(\beta) > 0$ and $A = \infty$ if $\mathfrak{Re}(\beta) < 0$. Integrating with respect to β and changing order of integration gives

$$F(\beta,x) - C_1 = \int_A^x \frac{y^{\beta-1}dy}{\log y} = \int_{A^\beta}^{x^\beta}\frac{dz}{\log z} = \mathrm{Li}(x^\beta)$$

letting β go to $\pm i\infty$ (depending on the sign of $Im(\beta)$) and $-\infty$ when $A = 0$ and $A = \infty$ respectively, one can show that the constant $C_1 = 0$.
Indeed note that as $|\beta| \to \infty$, $\log(1-\frac{s}{\beta}) \to 0$, hence $F(\beta,x) \to 0$. Now let $\beta = a + ib$, with $a, b \geq 0$ (the case $b < 0$ is the same), and consider the integral

$$\int_0^x \frac{y^{\beta-1}}{\log y}dy = \int_0^x \frac{e^{a\log y}e^{ib\log y}}{y\log y}dy = \int_{-\infty}^{\log x}\frac{e^{az}e^{ibz}}{z}dz.$$

Standard results in Fourier analysis indicate that as b goes to infinity the speed at which the integral swaps it sign increase and the cancellation effect will imply that the integral goes to 0. Similarly if $a < 0$ then clearly we have

$$\int_\infty^x \frac{y^{\beta-1}}{\log y}dy = \int_\infty^x \frac{e^{a\log y}e^{ib\log y}}{y\log y}dy = \int_0^{\log x}\frac{e^{az}e^{ibz}}{z}dz \xrightarrow{a\to-\infty} 0.$$

So indeed $C_1 = 0$, and in the cases of $\log(s-1)$ and $\log(1-\frac{s}{\rho})$ we get $\mathrm{Li}(x)$ and $\mathrm{Li}(x^\rho)$ respectively. For the terms coming from the Gamma function we can compute

$$\sum_n \int_\infty^x \frac{y^{-2n-1}}{\log y}dy = -\int_x^\infty \frac{\frac{y^{-2}}{1-y^{-2}}}{y\log y}dy = -\int_x^\infty \frac{dy}{y(y^2-1)\log y}.$$

Thus, putting all of the above together, pairing the roots ρ and $1-\rho$ of ξ and denoting them by $\rho = \frac{1}{2} + \alpha i$, $1 - \rho = \frac{1}{2} - \alpha i$ we get[5]

$$J(x) = \mathrm{Li}(x) - \sum_{\alpha} (\mathrm{Li}(x^{\frac{1}{2}+\alpha i}) + \mathrm{Li}(x^{\frac{1}{2}-\alpha i})) + \int_x^\infty \frac{dy}{y(y^2-1)\log y} + \log \xi(0).$$

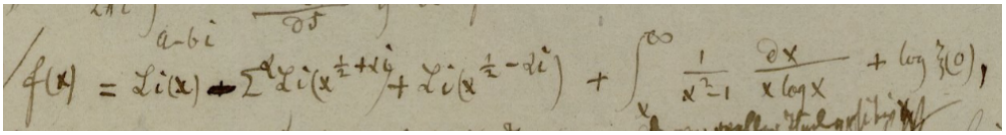


Figure 7: Rimann's main result, an explict formula for $J(x)$

Now $\pi(x)$ can be extracted from the formula of $J(x)$ using Möbius inversion formula:

$$J(x) = \sum_n \frac{1}{n} \pi(x^{\frac{1}{n}}) \Longleftrightarrow \pi(x) = \sum_{n \text{ not divisible by a square other than 1}} (-1)^{\mu(n)} J(x^{\frac{1}{m}})$$

Now, the terms $\log \xi(0)$ and $\int_x^\infty \frac{1}{y(y^2-1)\log y}$ are negligible as x goes to infinity. The terms $\sum \mathrm{Li}(x^\rho)$ are a lot more mysteries, Riemann refers to them as the periodic terms[6]. ignoring them for the moment and focusing on the main term $\mathrm{Li}(x)$ we get Riemann's approximation for $\pi(x)$[7]:

$$\pi(x) \sim \mathrm{Li}(x) - \frac{1}{2}\mathrm{Li}(x^{\frac{1}{2}}) - \frac{1}{3}\mathrm{Li}(x^{\frac{1}{3}}) - \frac{1}{5}\mathrm{Li}(x^{\frac{1}{5}}) + \frac{1}{6}\mathrm{Li}(x^{\frac{1}{6}}) - \frac{1}{7}\mathrm{Li}(x^{\frac{1}{7}}) \ldots$$

As the above discussion indicate, the main error term of the above formula will come from the periodic part $\sum \mathrm{Li}(x^{\frac{1}{\rho}})$. Note that $|\mathrm{Li}(x^\rho)| \leq \mathrm{Li}(x^{\mathfrak{Re}(\rho)})$ hence bounding the real value of zeros of ξ (equivalently of ζ) gives some bounds on those error terms, for example if the Riemann hypothesis is true then all those terms are bounded by $\mathrm{Li}(x^{\frac{1}{2}})$.

[5]where $\xi(0) = -\zeta(0) = \frac{1}{2}$

[6]He writes "In any future count it would be to keep track of the influence of the individual periodic terms in the expression for the density of the prime numbers"

[7]Riemann says "the known approximation $\pi(x) \sim \mathrm{Li}(x)$ is correct only up to terms of order $x^{\frac{1}{2}}$ and gives a value which is slightly too large". As it turns out, the conjecture $\pi(x) < \mathrm{Li}(x)$, even though supported by all numerical evidence, is false, as shown by Littlewood.

8 From Riemann's work to the prime number theorem

The well known prime number theorem (PNT) is the assertion that

$$\lim_{x\to\infty} \frac{\pi(x)}{\mathrm{Li}(x)} = 1$$

First conjectured by Gauss, it was long regarded as the most important open problem in number theory. Riemann's approach, though not a direct proof, played a crucial role in the eventual solution and brought mathematics remarkably close to the Prime Number Theorem.

Indeed, by Riemann's explicit formula, in order to prove the PNT one needs to show that the periodic terms are negligible compared to $\mathrm{Li}(x)$. As indicated above, if $\mathfrak{Re}(\rho) < 1$ for all roots ρ of ξ then $\frac{\mathrm{Li}(x^\rho)}{\mathrm{Li}(x)} \xrightarrow{x\to\infty} 0$. Then using uniform convergence one can try to exchange the limit and the sum to deduce that $\sum_\rho \frac{\mathrm{Li}(x^{\frac{1}{\rho}})}{\mathrm{Li}(x)} \xrightarrow{x\to\infty} 0$. Unfortunately, the above sum does not converge uniformly, and thus this approach fails. Nevertheless, this is not a fundamental obstacle: by using a slightly modified prime-counting function, one can obtain a similar explicit formula in which the periodic terms do converge uniformly. The modern approach is to work with the Chebyshev's ψ function (not to be confused with Jacobi ψ function) that sums up $\log p$ for every prime power p^k less than x:

$$\psi(x) = \sum_{p^k \le x} \log p.$$

von Mangoldt in 1895 used similar technics as the ones used by Riemann to give the explicit formula:

$$\psi(x) = x - \sum \frac{x^\rho}{\rho} - \log(2\pi) - \frac{1}{2}\log(1 - x^{-2}).$$

In 1896 Hadamard and Poussin independently proved the prime number theorem using the above approach. They used slightly different explicit formulas; Poussin specifically used

$$\int_0^x \frac{\psi(t)}{t^2} dt = \log x - \sum \frac{x^{\rho-1}}{\rho(\rho-1)} - \sum_n \frac{x^{-2n-1}}{2n(2n+1)} + \frac{\zeta'(0)}{x\zeta(0)} + const.$$

while Hadamard used

$$\int_0^x \frac{\psi(t)}{t} dt = x - \sum \frac{x^\rho}{\rho^2} - \sum_n \frac{x^{-2n}}{(2n)^2} - \frac{\zeta'(0)}{\zeta(0)} \log x + const.$$

The $\frac{1}{\rho^2}$ and $\frac{1}{\rho(\rho-1)}$ terms in the above guarantee that the sum converges uniformly. Thus, to show that all the terms besides the main one are negligible, one needs only to show that $\mathfrak{Re}(\rho) < 1$. From there, deducing the PNT is relatively simple, and indeed the main difficulty they had to overcome was showing $\mathfrak{Re}(\rho) < 1$ for all ρ.

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